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LLM-as-a-Judge: Direct Assessment, Pairwise Comparison & the Biases That Skew Both

Choose the right judging strategy for your task, and recognise the three biases that make a judge quietly wrong.

For engineers and researchers who need to evaluate large volumes of generated text without labelling every output by hand · 27 July 2026

Every section, key point, term and question from the full lesson is here. The extended explanations and the additional examples are not.

[Watch the source video](#)

[Read the full lesson](#)

[Read the highlights](#)

BEFORE YOU START

- Familiarity with prompting a language model
- A rough sense of why evaluating generated text is harder than evaluating a classifier

Overview

You have a thousand model outputs — chatbot replies, summaries, generated answers — and you need to know which are good. Labelling them by hand is accurate and impossibly slow. Traditional metrics like BLEU and ROUGE are fast but measure word overlap against a reference answer, which you often do not have and which says little about whether a summary is actually clear.

LLM-as-a-judge fills that gap: a language model evaluates other models' outputs. This lesson covers the two strategies for doing it — direct assessment against a rubric, and pairwise comparison between two candidates — along with what each is good for and how real users choose between them.

It then covers the part that decides whether your judge is trustworthy: the biases. Like human reviewers, LLM judges have systematic blind spots. They can favour whichever answer appeared first, prefer longer answers regardless of content, and rate their own output more highly than others'. None of these makes the approach unusable, but each one produces plausible-looking scores that are wrong in a consistent direction, so you need to know how to detect and correct for them.

KEY TAKEAWAYS

- ✓ LLM-as-a-judge enables reference-free evaluation: judging quality without a gold-standard answer to compare against.
- ✓ Direct assessment scores each output independently against a rubric you design; pairwise comparison asks which of two outputs is better.
- ✓ With multiple candidates, pairwise judgements can be combined by a ranking algorithm to produce an overall ordering.
- ✓ In IBM's user research on the open-source EvalAssist framework, about half preferred direct assessment for its clarity and control, a quarter preferred pairwise for subjective tasks, and the rest combined both.
- ✓ The three advantages over traditional metrics are scale, flexibility as your criteria evolve, and the ability to judge nuanced qualities like whether text sounds natural.
- ✓ Positional bias: the judge favours an output because of where it appeared, not because it is better.
- ✓ Verbosity bias: the judge treats length as a proxy for quality and prefers longer answers even when they ramble.
- ✓ Self-enhancement bias: the judge favours output it recognises as its own.
- ✓ Positional swaps — running the same comparison with the order reversed — detect positional bias directly. Bias means stay vigilant, not that the system is broken.

01 Why traditional metrics run out

0:00

Manual labelling does not scale and word-overlap metrics need a reference answer they cannot always have.

KEY POINTS

- Manual labelling of hundreds of outputs is accurate but does not scale.
- BLEU and ROUGE measure word overlap against a reference answer, which is often unavailable.
- Overlap metrics cannot capture qualities like naturalness, coherence, or whether text sounds human.
- Reference-free evaluation judges an output on its own merits, with no gold answer required.

Where word overlap misleads

Two summaries of the same article. One reuses most of the reference's vocabulary but scrambles the causal order into something incoherent; the other paraphrases cleanly and reads well. Overlap metrics score the first higher. A human — or a judge model — scores the second higher, because the question was never about vocabulary.

02 Strategy one: direct assessment

0:42

Design a rubric and score each output against it independently, one at a time.

KEY POINTS

- Design a rubric: the criterion, plus the options the judge may return.
- Each output is scored independently, producing a standalone label.
- The strength is clarity and control — you define exactly what is measured.
- Well suited to compliance-style checks where the standard is fixed and external.
- Standalone labels aggregate naturally into pass rates across a large batch.

A direct assessment rubric

The rubric is short by design. One criterion, explicit options, and the output judged on its own terms.

Criterion: Is this summary clear and coherent?

Options:

yes – the summary is clear

no – the summary is not clear

<summary>{output}</summary>

Return one option and explain your reasoning.

03 Strategy two: pairwise comparison

1:23

Instead of a standalone label, ask which of two outputs is better — then rank many

outputs from the accumulated comparisons.

KEY POINTS

- The judge is asked which of two outputs is better rather than for a standalone label.
- Relative judgements are easier and more consistent than absolute ones for subjective qualities.
- A ranking algorithm converts many pairwise verdicts into an overall ordering of all candidates.
- The limitation: a pairwise win is relative and says nothing about whether either output met your bar.

A pairwise comparison prompt

The judge sees both candidates and picks one. Note that the ordering of A and B is itself a variable you will need to control for.

Criterion: clarity

Which of these outputs is clearer?

<option_A>{output_a}</option_A>

<option_B>{output_b}</option_B>

Return A or B, and explain your reasoning.

04 Which strategy do people actually choose?

2:06

User research on the open-source EvalAssist framework found the split was roughly half direct assessment, a quarter pairwise, and the rest a combination — with the choice

depending on both task and user.

KEY POINTS

- About half of participants preferred direct assessment for its clarity and rubric control.
- About a quarter preferred pairwise comparison, particularly for subjective tasks.
- The remainder combined both: direct assessment for compliance, then pairwise ranking to pick the best.
- The combined approach gives an absolute quality floor plus a relative ordering above it.
- The conclusion was that the choice is task- and user-dependent, not universal.

The two-stage combined pipeline

Filter on an absolute standard first, then rank what survives. This is how the combined-approach participants used the two strategies together.

```
# Stage 1 - direct assessment as a compliance gate
passing = [o for o in outputs
            if judge_direct(o, criterion="meets policy") == "yes"]

# Stage 2 - pairwise ranking to select the best survivor
best = rank_pairwise(passing, criterion="clarity")[0]
```

05 Why use an LLM judge: scale, flexibility, nuance

2:47

It handles volume, it adapts as your criteria shift, and it can evaluate qualities no overlap metric reaches.

KEY POINTS

- Scale: handles hundreds or thousands of outputs across many models and prompts, returning structured feedback quickly.
- Flexibility: criteria naturally shift as you see more data, and refining a prompt is far cheaper than reimplementing a metric.
- Nuance: subjective, reference-free questions like 'does this sound human?' become measurable.
- Criteria refinement is an expected part of the process, not a sign you got the rubric wrong.

Criteria refinement in practice

Your first rubric asks whether a summary is clear. After reading eighty judged outputs you notice that the failures are nearly all about ordering — the facts are right but the sequence is confusing. You refine the criterion to name that explicitly, which no fixed metric would have let you do.

v1

Criterion: Is this summary clear and coherent?

v2, after seeing where it actually failed

Criterion: Does this summary present events in an order a reader can follow, with each claim's context established before it is used?

06 The three biases that skew a judge

3:29

Positional, verbosity, and self-enhancement bias each make a judge favour outputs for reasons unrelated to quality.

KEY POINTS

- Positional bias: favouring an output because of its position, e.g. always choosing option A regardless of content.
- Verbosity bias: favouring longer output regardless of quality, because length reads as thoroughness.
- Self-enhancement bias: favouring output the model recognises as its own.
- These biases skew results systematically rather than adding random noise, so they do not average out across a large sample.
- Bias does not mean the system is broken — it means you stay vigilant and design checks for it.

Verbosity bias in a rubric

If your judge keeps approving rambling answers, the fix is often to name the failure in the rubric rather than to abandon the judge. Say explicitly that length is not quality.

Criterion: helpfulness

Judge only whether the customer's question was answered.

Do NOT reward length. A short, complete answer is better than a long one that repeats itself or adds unrequested detail.

An answer over 120 words that could have been given in 40 should be marked down for verbosity.

07 Catching bias with positional swaps

4:54

Run the same comparison twice with the options in reverse order; if the verdict flips, the judgement was driven by position rather than content.

KEY POINTS

- Run each comparison twice with the options in reversed order.
- A consistent judge picks the same output both times; a positionally biased one keeps picking the same slot, flipping the verdict.
- Per-pair, a flipped verdict marks that comparison as unreliable.
- In aggregate, the flip rate quantifies how positionally biased the judge is on your task.
- The general method: vary only the suspected factor, hold content constant, and check whether the verdict moves.

Implementing a positional swap check

Both orderings are judged and only agreeing verdicts are kept. The disagreement rate is itself a useful metric to track.

```
def robust_compare(judge, x, y, criterion):  
    """Judge both orderings; return a winner only if they agree."""  
    first = judge(a=x, b=y, criterion=criterion) # x in slot A  
    second = judge(a=y, b=x, criterion=criterion) # x in slot B  
  
    winner_1 = x if first == "A" else y  
    winner_2 = y if second == "A" else x  
  
    if winner_1 is winner_2:  
        return winner_1 # consistent - trust it  
    return None # flipped - positional bias  
  
results = [robust_compare(judge, a, b, "clarity") for a, b in pairs]  
flip_rate = results.count(None) / len(results)  
print(f"positional bias flip rate: {flip_rate:.0%}")
```

Glossary

LLM-as-a-judge

Using a language model to evaluate the outputs of another model, applying a criterion you specify.

Reference-free evaluation

Judging an output on its own merits, without needing a gold-standard answer to compare it against.

Direct assessment

A judging strategy where each output is scored independently against a rubric you designed.

Pairwise comparison

A judging strategy where two outputs are shown together and the judge picks which is better.

Rubric

The criterion being evaluated plus the set of options the judge may return.

Ranking algorithm

A method for converting many individual pairwise verdicts into an overall ordering of all candidates.

BLEU / ROUGE

Traditional text metrics that score an output by how much its wording overlaps with a reference answer.

Positional bias

A judge favouring an output because of the position it appeared in rather than its content.

Verbosity bias

A judge favouring longer outputs regardless of quality, treating length as a signal of thoroughness.

Self-enhancement bias

A judge rating its own generated output more highly than others', independent of actual quality.

Positional swap

Re-running a pairwise comparison with the two options in reverse order to test whether the verdict was driven by position.

EvalAssist

IBM's open-sourced evaluation framework, used in the user research on how people choose between judging strategies.

Check yourself

Answer first, then open each one.

You need to check whether a thousand generated replies each meet a fixed regulatory standard. Which strategy fits, and why?

Direct assessment. The standard is absolute and external, so each output must be judged on its own against a written rubric. Pairwise comparison would only tell you which reply is better than another, which is useless here — both could fail the standard and one would still win the comparison.

Why does positional bias not average out if you simply run a large number of comparisons?

Because it is systematic rather than random. A judge that consistently prefers slot A pushes every comparison in the same direction, so more samples produce a more confident wrong answer rather than converging on the truth. Random noise averages out; a consistent lean does not.

How does a positional swap distinguish a genuine quality judgement from a positional artefact?

Because it holds content constant and changes only position. A judge responding to content will select the same output in both orderings — the winner just moves slot. A judge responding to position keeps picking the same slot, so the winner changes. Disagreement between the two runs isolates position as the cause.

Why is the flexibility of an LLM judge described as an advantage over traditional metrics rather than a lack of rigour?

Because criteria genuinely do shift as you see more data — you discover that what you care about is more specific than what you first wrote down. A fixed metric forces your evaluation to keep measuring the original approximation, while a prompt-based judge lets the measurement track your improved understanding. The rigour comes from validating the judge, not from freezing the criterion.

A judge scores summaries from three different models and consistently rates one model's output highest. What should you check before concluding that model is best?

Whether the judge is the same model that produced the favoured output, which would point to self-enhancement bias. You should also check for verbosity bias by comparing average output lengths — if the favoured model simply writes longer answers, the judge may be rewarding length. Validating against human labels settles it.

Where to go next

- Take an existing pairwise evaluation and add a positional swap, then measure your judge's flip rate — treat anything above a few percent as a problem to fix.
- Run the padding probe on your own judge: score a concise answer, then score it again with filler added, and see whether the score moves.

- Rewrite one of your rubrics to explicitly penalise verbosity, and compare the label distribution before and after.
 - Explore the open-source EvalAssist framework to see how a purpose-built tool structures rubrics and bias checks.
 - Build the two-stage pipeline: direct assessment as a compliance gate, then pairwise ranking to select the best output among those that pass.
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Lesson generated from a YouTube transcript with YouTube Lesson Builder.

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